Course Project Check

Due: Sunday, October 25, 2026 at 11:59PM, in Canvas. Worth 5 points.

Purpose

The purpose of this checkpoint is to make sure you have a clear and well-defined understanding of the data you intend to use for the course project (project.html) and the statistical question you plan to investigate. Getting these two things right now saves you from discovering in December that your data cannot answer your question.

What to submit

Upload a short description of the dataset you have selected for your project (**1 page maximum**) to Canvas. It must address the three points below.

1. **The dataset (2 points).** Give the source of the data: a **working link** to the dataset, plus a citation (who collected or published it, and when). Then briefly describe the variable(s) that interest you: what each one measures, its units, and whether it is **quantitative** or **categorical**.
2. **Problems with the data (1 point).** Highlight any potential challenges or limitations of the data, such as missing values, outliers, data quality issues, or limits on who or what the data describe. For each one, say **how you plan to handle it**.
3. **Your statistical question (2 points).** Clearly state the specific statistical question you plan to address in your project. A good question:
    - is about the variable(s) you described in point 1, and can be answered with the data you actually have;
    - names the group (population) it is about;
    - asks about a **summary** such as a mean, a proportion, or a difference between groups, rather than about a single individual;
    - could be answered with a method from this class (for example, a test for a mean or a proportion, a comparison of two groups, or a χ^2 test).

You do not need to draw your sample, compute any statistics, or write hypotheses in symbols yet. Those are Steps 3–5 of the project.

Grading

Component	Full credit	Partial credit	Points
1. Dataset and source	A real dataset, a working link and a citation; the variable(s) of interest described, with type and units	1 pt: dataset identified, but the link is missing or broken, or the variables are not described	2
2. Problems with the data	At least one real problem with **this** dataset, and a plan for dealing with it	0.5 pt: a problem is named but there is no plan, or it is generic (“data can have errors”)	1
3. Statistical question	One specific question, answerable with this dataset’s variables and a method from class, naming its population	1 pt: the question is vague, is about a single individual, or the data cannot answer it	2
Total			**5**

Common ways to lose points:

- Linking to a search page, a website’s home page, or a paper behind a paywall instead of to the data itself. Click your link from a private browser window before you submit.
- Naming a topic instead of a question (“I want to look at sleep and grades”).
- Asking a question your dataset cannot answer, such as asking about income when the dataset has no income variable, or asking about a change over time in data from a single year.
- Writing “there are no problems with this data.” Every real dataset has limits, even if it is only that it describes one place or one period of time.

If you use AI in any capacity, the AI Use (project.html#ai-use) rules for the project apply to this checkpoint too.

Full-credit example

INSTRUCTOR NOTE

Below is an example of a submission that would earn **5 out of 5**. The notes in *italics* explain why each part earns its points and are not part of the submission. It fits within the **1 page limit**. **Do not use this dataset or question for your own project.** The example was prepared with the help of Claude (Anthropic), and every number in it was checked against the data file.

Project check: body mass of male and female Gentoo penguins

1. The dataset

I will use the **Palmer Penguins** dataset, which records body measurements of 344 adult penguins of three species (Adélie, Chinstrap and Gentoo) nesting on three islands near Palmer Station, Antarctica, during the 2007, 2008 and 2009 seasons. The data were collected by Dr. Kristen Gorman and the Palmer Station Long Term Ecological Research (LTER) program.

- **Link to the data:** https://allisonhorst.github.io/palmerpenguins/ (the data file itself is penguins.csv (https://raw.githubusercontent.com/allisonhorst/palmerpenguins/main/inst/extdata/penguins.csv)).
- **Citation:** Horst, A. M., Hill, A. P., & Gorman, K. B. (2020). *palmerpenguins: Palmer Archipelago (Antarctica) penguin data* (R package version 0.1.0) [Data set]. https://doi.org/10.5281/zenodo.3960218. Originally published in Gorman, K. B., Williams, T. D., & Fraser, W. R. (2014). Ecological sexual dimorphism and environmental variability within a community of Antarctic penguins (genus *Pygoscelis*). *PLoS ONE, 9*(3), e90081. https://doi.org/10.1371/journal.pone.0090081

I will only use the **124 Gentoo penguins**, all of which were measured on Biscoe Island. The two variables I am interested in are:

- `body_mass_g` , the penguin's body mass in grams. This is **quantitative**.
- `sex` , whether the penguin is female or male. This is **categorical**, with two categories.

2. Problems with the data

- **Missing values.** Five of the 124 Gentoo penguins have no value for `sex` , and one of those five also has no body mass. I cannot put these penguins in either group, so I will

remove them before I draw my sample. That leaves 119 penguins. Since only 5 are removed, this should not change the results much, but I will mention it in my report.

- **Limited scope.** All of the Gentoo penguins come from one island and three nesting seasons (2007–2009). My conclusions will only describe Gentoo penguins nesting on Biscoe Island in those years, not Gentoo penguins everywhere or today.
- **Possible outliers.** Body mass can be affected by an unusually large or small bird. Once I draw my sample, I will check for outliers with the $1.5 \times IQR$ rule and report whether any outliers change my conclusion.

3. My statistical question

On average, do male Gentoo penguins nesting on Biscoe Island weigh more than female Gentoo penguins?

I will answer this by comparing the mean body mass of the males and the females in my sample.

Why it earns full credit:

- *Dataset (2/2): there is a working link straight to the data file, a full citation, and each variable is described with its units and its type.*
- *Problems (1/1): every problem is specific to this dataset (the five penguins with missing sex are counted), and each one comes with a plan for handling it.*
- *Question (2/2): it is one question, it names its population (Gentoo penguins on Biscoe Island), it asks about a summary (the mean body mass of each sex), and the two variables in the dataset are exactly what is needed to answer it with a comparison of two means.*